Internet of Things (IoT)



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Outline

- Discussion
- Genesis of IoT
- What is IoT ?
- IoT Devices
- Importance of Communication in IoT
- Importance of IoT for Computing Undergraduates
- References



DISCUSSION



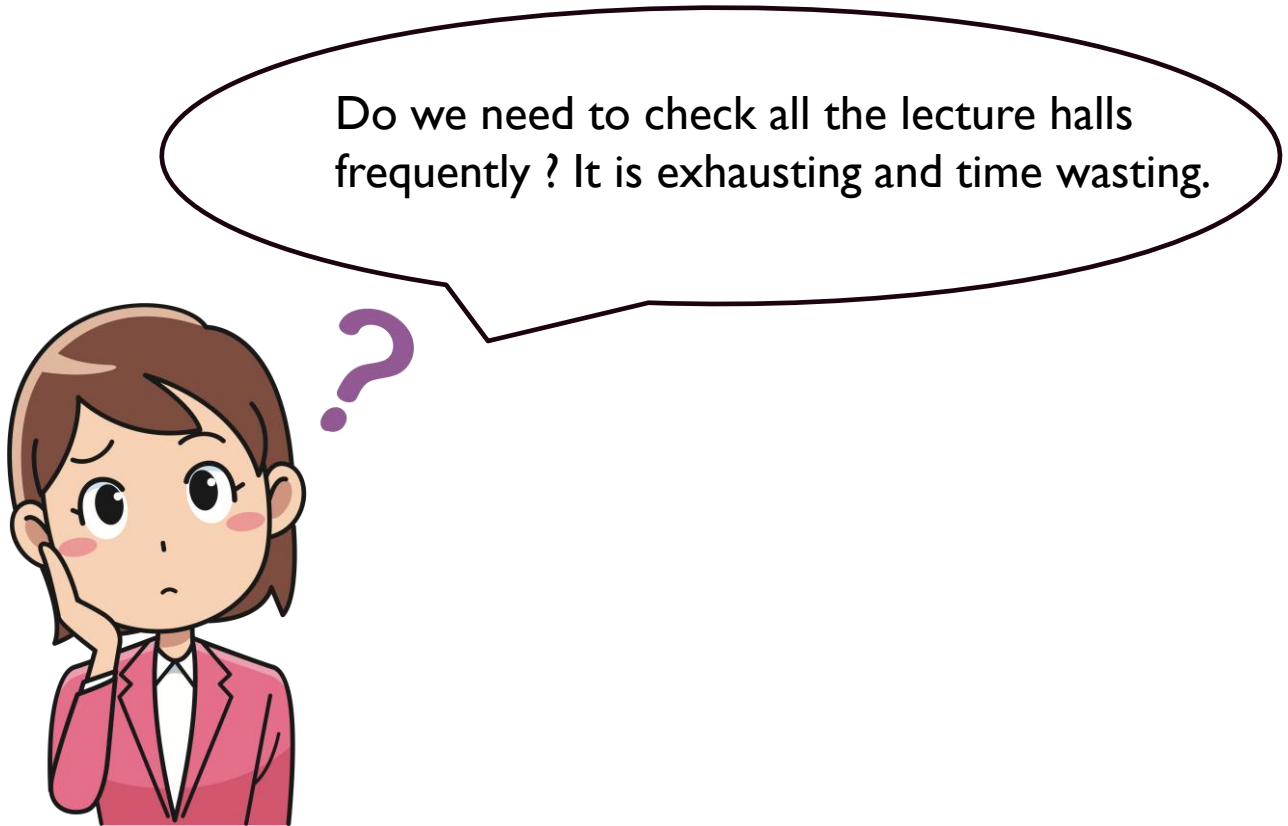
The “Lost Keys” Scenario

“Last week I was locked out of my house because I forgot my keys inside. I had to wait 2 hours for a locksmith.”



The “AC Wastage” at NSBM

At our campus, many classrooms have the AC running even when nobody is inside.



The “Forgotten Medicine” Concern (Elderly Care)

My grandmother lives alone, and she sometimes forgets to take her medicine.



The “Water Tank Overflow” Problem

Last night, I was so tired I forgot to turn off the water pump. When I woke up, the tank had overflowed, and the yard was flooded.



I will have to work extra hours to pay this electricity bill. I have no time to clean the yard today.

I can`t bear this anymore. I am going to hire someone to manage this ? How much do I need to pay the worker then ?

Group Activity

Think of 2–3 real-life problems you've personally experienced or observed at home, in college, or in public that made you think, 'There should be an easier way to do this.'

Example Prompts

- "I always forget to lock the gate."
- "Our canteen food finishes before lunch break ends"



IOT BASED SYSTEMS
VS
NON IOT SYSTEMS



Home Water Tank with No IoT



Home Water Tank with No IoT



- Water tank is manually monitored
- Manual ON/OFF switch
- No alerts – you might forget to refill
- You have to check water level and turn pump ON/OFF
- No data recorded
- No remote access – you must be near the tank

Home Water Tank with No IoT





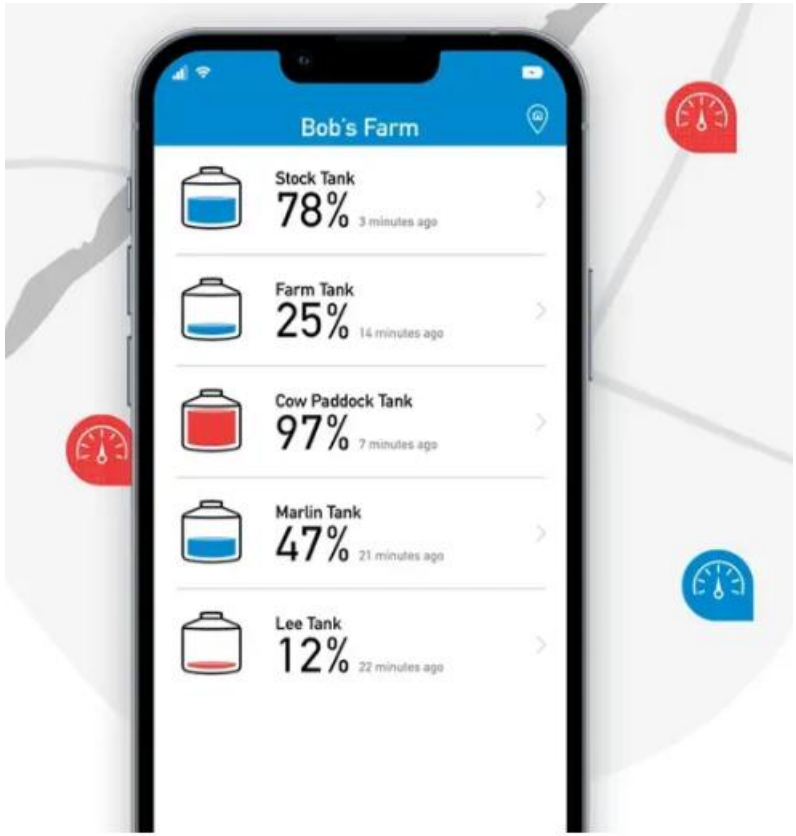
Home Water Tank with IoT



- Tank automatically monitored and managed
- System works automatically or with a tap on your phone
- control from anywhere using the internet

Home Water Tank with IoT





- Sends alerts when water level is low or pump is ON too long
- Pump can turn ON/OFF automatically based on sensor data
- Water usage is recorded for analysis and future planning
- Saves time, water, and reduces human error

Key Differences



Non IoT



IoT

- Dumb system
- Isolated
- Human-dependent
- Hard to scale
- Reactive Maintenance

- Smart system
- Connected
- Can be automated with logic or AI
- Easy to connect many devices
- Maintenance is predictive with data insights

GENESIS OF IOT



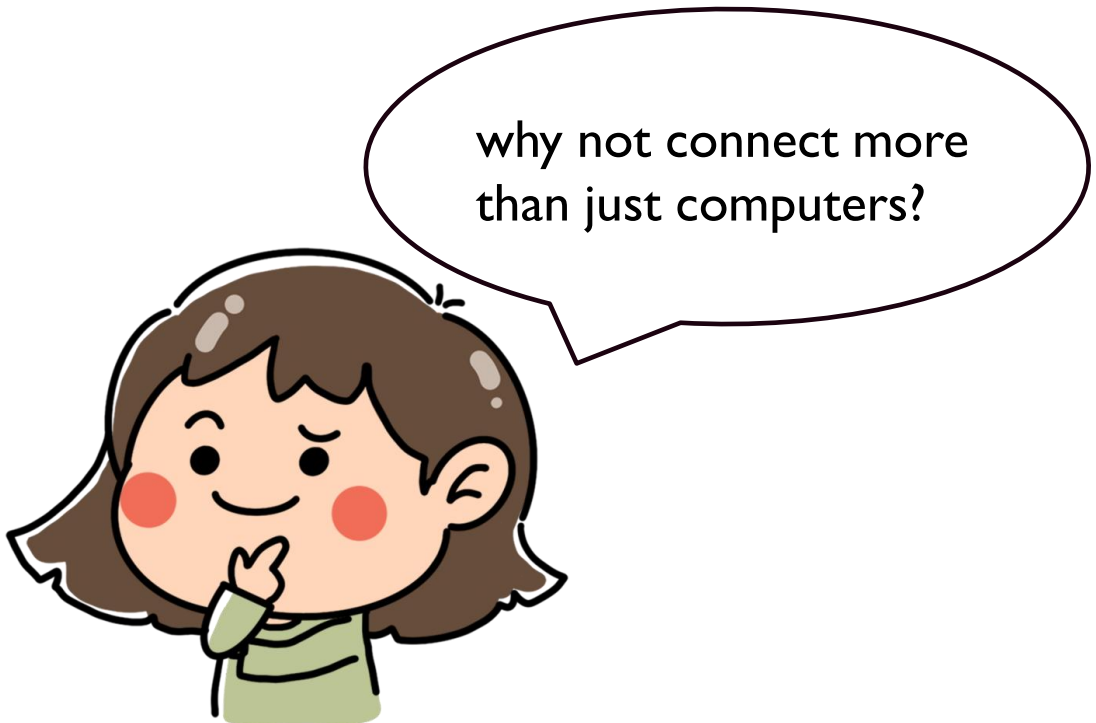
Life Before IoT

Before IoT, most electronic devices (TVs, washing machines, fans, etc.) were *not connected* to the internet or each other.
They worked **independently** and required **manual operation**.



The Internet Boom (1990s)

The **internet** started connecting **computers globally**, allowing **information sharing**, **emails**, and **web browsing**.
This gave birth to the **idea of connected systems.**



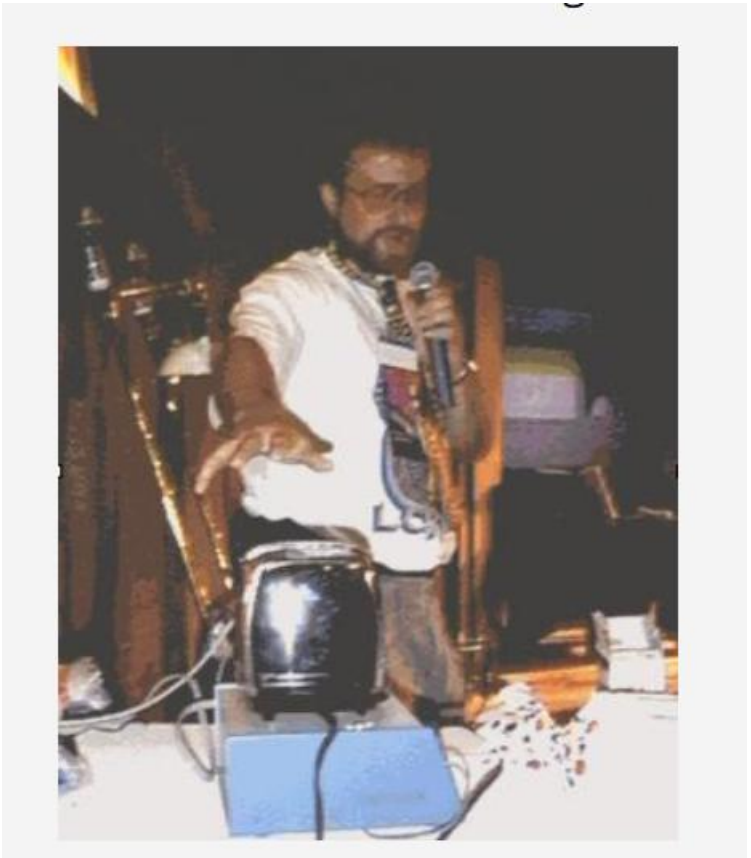


The First "Thing" Gets Online (1990)

In 1990, John Romkey created the first IoT-like device — a toaster controlled over the internet!

This was a simple but revolutionary idea:

- ❑ A **thing** (toaster)
- ❑ Connected to the **internet**
- ❑ Controlled remotely



Coining the Term "Internet of Things" (1999)

Kevin Ashton, a British technologist at MIT, coined the term **"Internet of Things"** while working on supply chain optimization using RFID.



1982 - Vending machine

The Coca-Cola vending machine story is often cited as one of the earliest real-world examples of an IoT-like system - even before the term *Internet of Things* was coined.

Place: Carnegie Mellon University (CMU), Pittsburgh, USA
Inventors: A group of computer science students and faculty

Problem: The vending machine was located **far from the computer lab**. Students would walk all the way there only to find:

Solution: They connected the **Coca-Cola vending machine** to the **ARPANET** (the early version of the internet), using a **networked computer**.

Significance: First known instance of a **physical object** being **monitored remotely** using a network



WHAT IS IOT ?



How does IoT transform our lives?

Life Simplified with Connected Devices

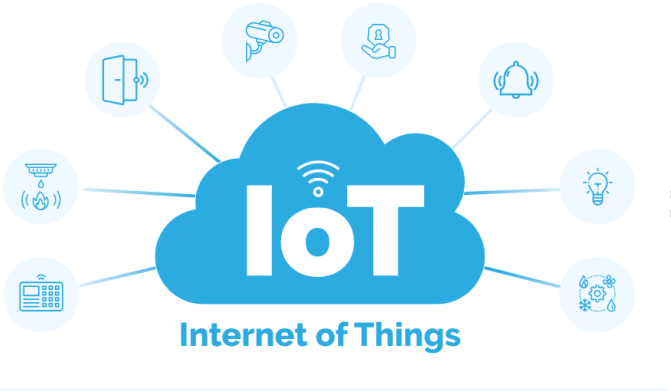


How does IoT transform our lives?

Future of IoT with Advance Technologies | Internet of Things



IoT Definition



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IoT **Internet** **Devices** **Communication**

IoT Definition



IoT

- Sense things
- send information
- receive commands
- make decisions

All without human help, using the internet.

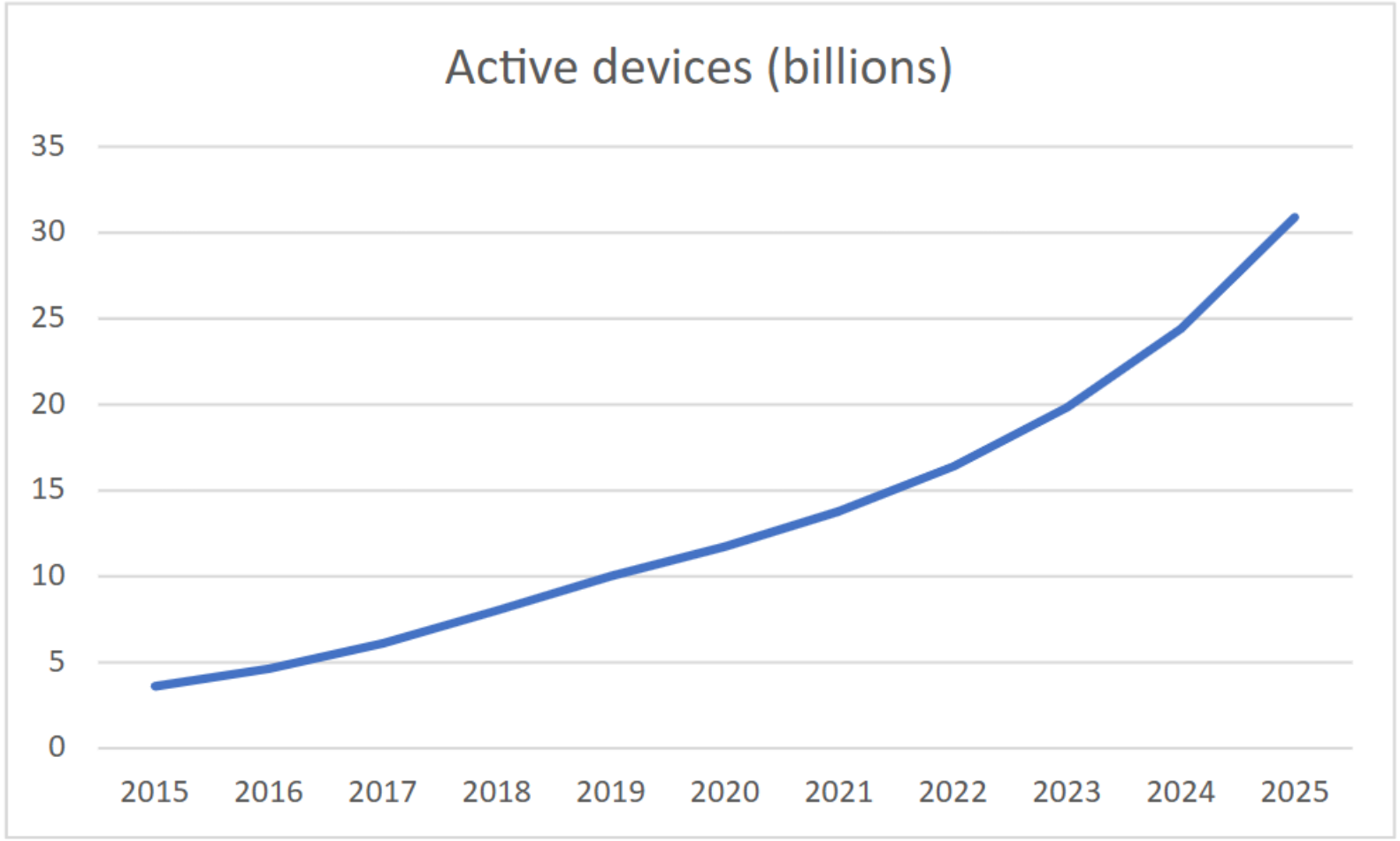
Active Devices in the World

It is estimated that by the end of 2020, **30 billion IoT devices** were deployed and connected to the Internet.

Looking to the future, it is estimated that by 2025, IoT devices will be gathering almost 80 zettabytes of data or **80 trillion gigabytes**.



Active Devices in the World



Why is IoT important ?



Smarter
(automatic)



Connected
(online)



Efficient
(saves time & energy)

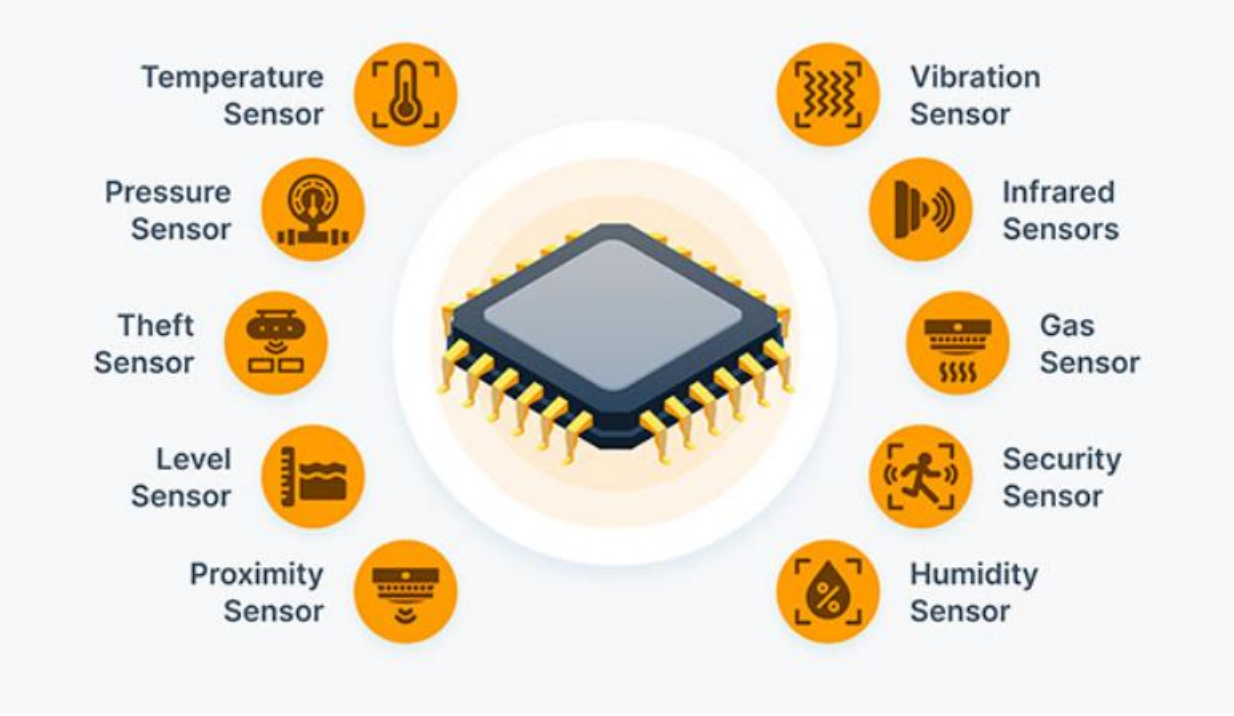
IOT DEVICES



Sensors

Sensors – The Ears & Eyes

Sensors gather information from the world, such as measuring speed, temperature or location.

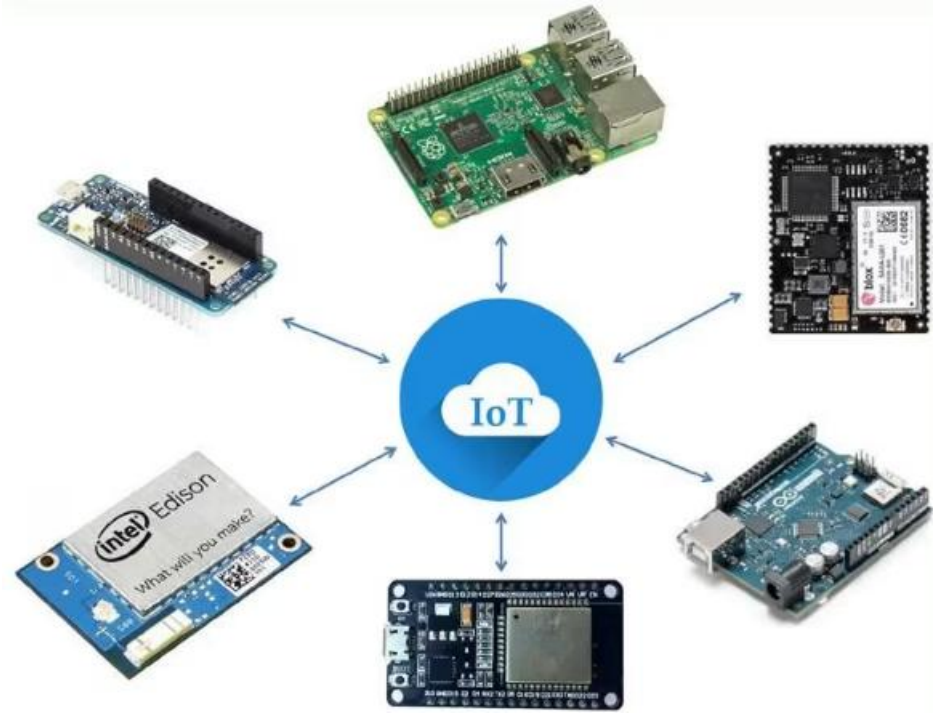


Microcontrollers

Microcontrollers – Brain



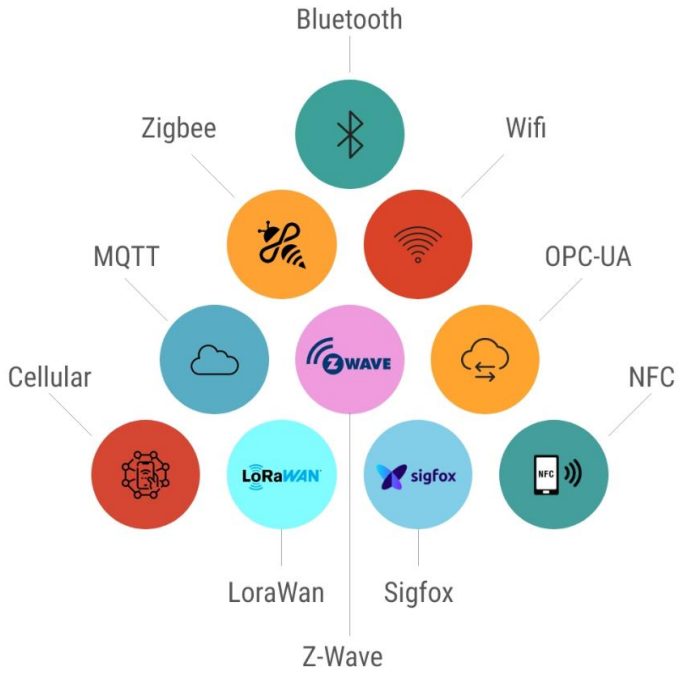
A microcontroller (also referred to as an MCU, short for microcontroller unit) is a small computer



Connectivity

Communication – Mouth & Ears

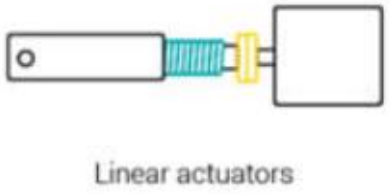
It helps the device **talk to the internet or your phone**.



Actuators

Actuators – The Hands/Tools

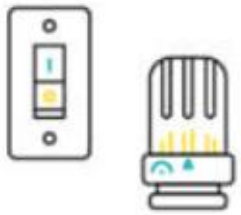
It **does something physical** based on a decision.



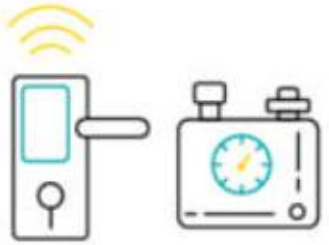
Linear actuators



Motors



Relays



Solenoids

Cloud

Cloud – The Memory and Dashboard

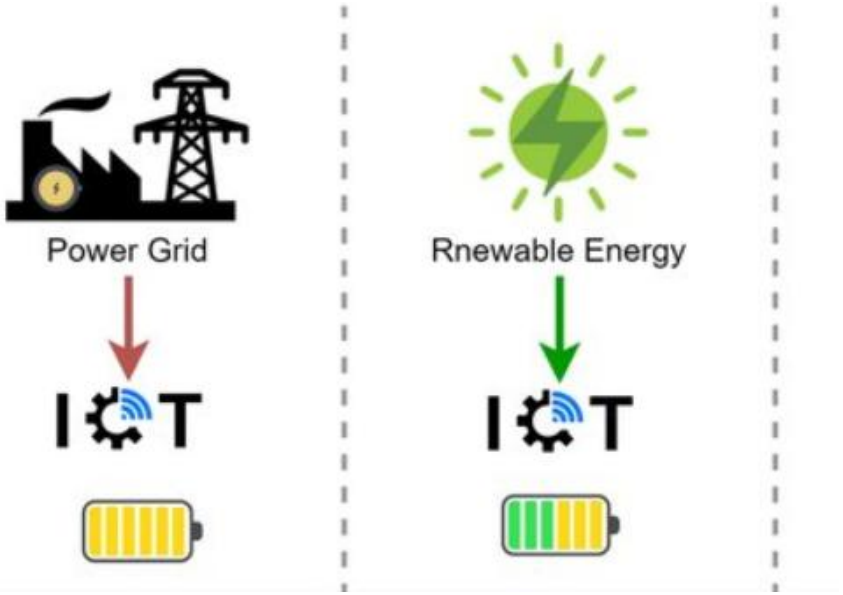
It **stores the data** and sometimes shows it in graphs.



Power Source

Power Source – The Heart

It gives **energy** to the whole system.





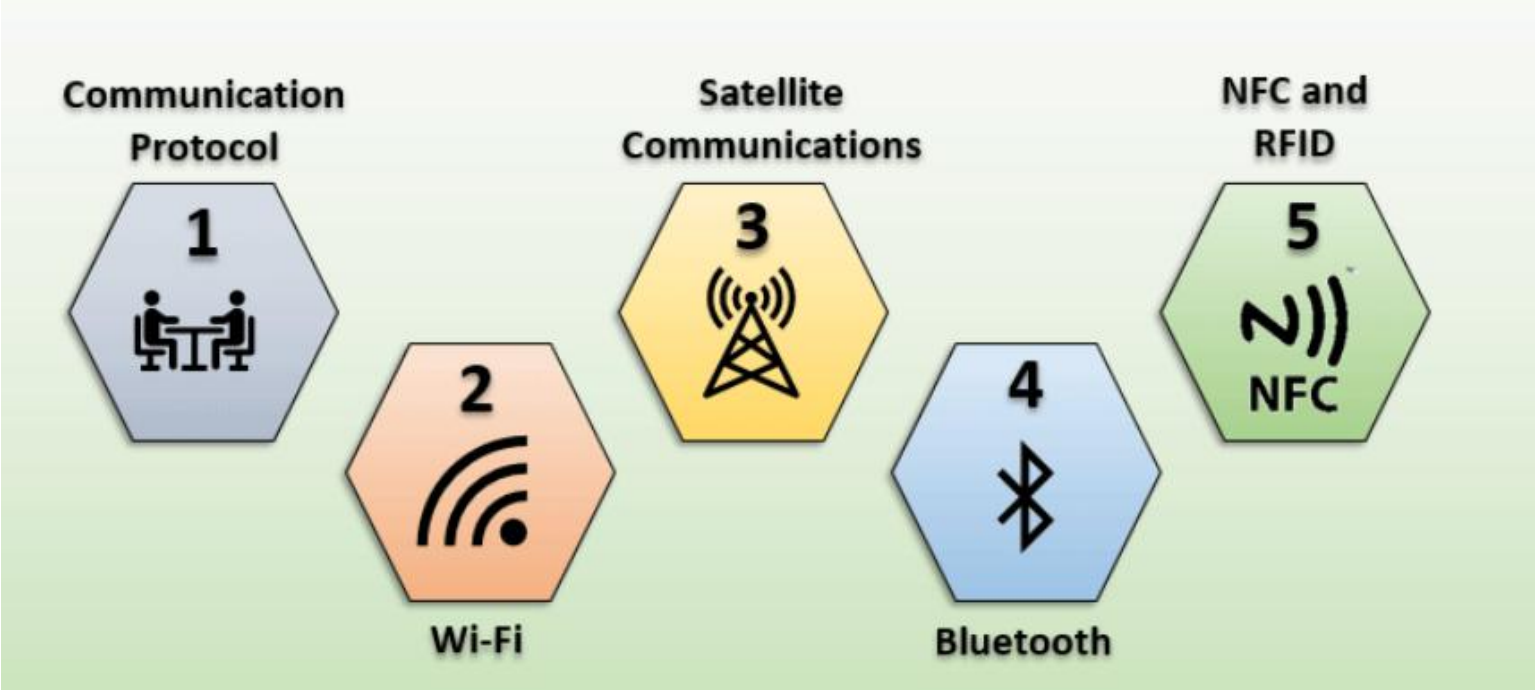


Group Activity



Brainstorm and Design IoT-based solutions to transform your campus into a **Smart Campus** by addressing **one** of the real-world problems listed in the slide.

IMPORTANCE OF COMMUNICATION IN IOT



Importance of Communication in IoT

The main idea behind IoT is:

- Devices **collect data** from the physical world (like temperature, motion, light, etc.)
- But that's not enough. To be useful, the data must be **sent to another device, a server, or a cloud** — to be stored, processed, or trigger actions.
- Without communication, an IoT device is like a person with no voice and no ears.
- In short: **No communication = No IoT**



IMPORTANCE OF IOT FOR COMPUTING UNDERGRADUATES



Importance of IoT for Computing Undergraduates

- ❑ IoT is a Major Part of the Future of Computing
  - ✓ You will **design** the systems, **write** the software, and **secure** the networks.

- ❑ Real-World Problem Solving
  - ✓ IoT is about solving **human problems using computing power**
  - ✓ Computing students will build apps, platforms, and analytics systems



- ❑ Full-Stack Development Opportunity
  - ✓ IoT includes **hardware + software + cloud + analytics**.

- ❑ Industry Demand
  - ✓ Almost every major tech company — from Google to Bosch to IBM — has **IoT divisions**.

Importance of IoT for Computing Undergraduates

- Data Science + AI + IoT = Smart Systems
  - ✓ IoT devices generate **continuous data streams** (sensor readings, GPS, logs).
  - ✓ Computing students turn **data into intelligence**.

- Cybersecurity and IoT
  - ✓ IoT systems are **highly vulnerable** — especially when poorly designed.
  - ✓ The future of **cybersecurity in IoT** depends on computing minds.

- Innovation, Startups, and R&D
  - ✓ IoT is an **entrepreneurial hotbed** — most modern smart devices started as student projects or hackathon ideas.



IoT Trends in 2025

What Are The Biggest IoT Trends Of 2025 And Why



Recommended Readings

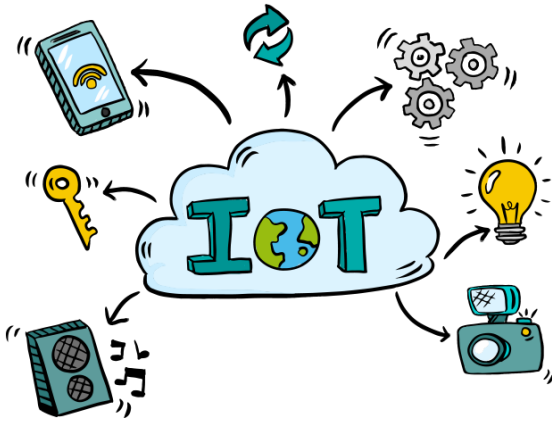


- IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017

- Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education

- The Fourth Industrial Revolution, by Klaus Schwab

Thank you!



Questions

